Probability Theory and Random Processes (MA225)

Lecture Slides

Lecture 31

Limiting Distributions

Def: For $i \in S$, $V_i^{(n)} = \#\{t : 0 \leq t \leq n, X_t = i\} = \sum_{k=0}^{n} \delta_i(X_k)$ is the number of visits to state i during $\{0, 1, \ldots, n\}$.

Def: For $i \in S$, define $L_i^{(n)} = \frac{V_i^{(n)}}{n+1}$. Then, $\left\{L_i^{(n)} : i \in S\right\}$ is called empirical distribution at time n.

Theorem: [Law of large numbers for MC] Let $\{X_n : n \geq 0\}$ be an irreducible and recurrent MC. Fix a state i. Suppose that $P(X_0 = i) = 1$. Then, for any state j

$$L_j^{(n)} \to \begin{cases} \frac{\gamma_j^i}{E_i(T_i)} & \text{if } E_i(T_i) < \infty \\ 0 & \text{if } E_i(T_i) = \infty \end{cases}$$

with probability 1. In particular, if i is positive recurrent, then $L_i^{(n)} \to \frac{1}{E_i(T_i)} = \pi_i$ with probability 1.

Theorem

Theorem: Fix the state $i \in S$. Then

1. i is transient iff $\sum_{k=0}^{\infty} p_{ii}^{(k)} < \infty$ (Already proved).
2. i is null recurrent iff $\sum_{k=0}^{\infty} p_{ii}^{(k)} = \infty$ and
$$\lim_{n\to\infty} \frac{1}{n+1} \sum_{k=0}^{n} p_{ii}^{(k)} = 0.$$
3. i is positive recurrent iff $\sum_{k=0}^{\infty} p_{ii}^{(k)} = \infty$ and
$$\lim_{n\to\infty} \frac{1}{n+1} \sum_{k=0}^{n} p_{ii}^{(k)} > 0.$$

Proof: Here, the proof is given assuming irreducibility of the MC.
Proof of (2) and (3): Suppose i is null recurrent. As i is recurrent, $\sum_{k=0}^{\infty} p_{ii}^{(k)} = \infty$. Now, by previous theorem,

$$\lim_{n\to\infty} L_i^{(n)} = 0 \implies \lim_{n\to\infty} E_i\left(L_i^{(n)}\right) = 0 \implies \lim_{n\to\infty} \frac{1}{n+1}\sum_{k=0}^{n} p_{ii}^{(k)} = 0.$$

Now, suppose that i is positive recurrent. As i is recurrent, $\sum_{k=0}^{\infty} p_{ii}^{(k)} = \infty$. Again, by previous theorem,

$$\begin{aligned}
&\lim_{n\to\infty} L_i^{(n)} = \frac{1}{E_i\left(T_i\right)} \\
&\implies \lim_{n\to\infty} E_i\left(L_i^{(n)}\right) = \frac{1}{E_i\left(T_i\right)} \\
&\implies \lim_{n\to\infty} \frac{1}{n+1}\sum_{k=0}^{n} p_{ii}^{(k)} = \frac{1}{E_i(T_i)} > 0.
\end{aligned}$$

Example

For a state i, what about $\lim_{n\to\infty} p_{ii}^{(n)}$?

Example 1: Consider the MC with the following TPM

$$\begin{array}{cc} & \begin{array}{cc} 0 & 1 \end{array} \\ \begin{array}{c} 0 \\ 1 \end{array} & \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{array}.$$

- This is an irreducible positive recurrent MC with unique stationary distribution $\left(\frac{1}{2}, \frac{1}{2}\right)$.
- But, here $p_{00}^{(2n-1)} = 0$ and $p_{00}^{(2n)} = 1$ for all $n \geq 1$.
- Thus, $\lim_{n\to\infty} p_{00}^{(n)}$ does not exist.

Period

Def: The period of a state i is defined by the greatest common divisor of all integers $n \geq 1$ for which $p_{ii}^{(n)} > 0$, *i.e.*,

$$d_i = \begin{cases} \gcd\left\{n \geq 1 : p_{ii}^{(n)} > 0\right\} & \text{if } \left\{n \geq 1 : p_{ii}^{(n)} > 0\right\} \neq \emptyset \\ 0 & \text{if } \left\{n \geq 1 : p_{ii}^{(n)} > 0\right\} = \emptyset. \end{cases}$$

Further, i is called aperiodic if $d_i = 1$.

Example 2: Consider the MC with the following TPM

$$\begin{array}{cc} & \begin{array}{cc} 0 & 1 \end{array} \\ \begin{array}{c} 0 \\ 1 \end{array} & \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{array}.$$

- $p_{00}^{(2n-1)} = 0$ and $p_{00}^{(2n)} = 1$ for all $n \geq 1$.
- Thus, $d_0 = 2$.

Example

Example 3: $S = \{0, \pm 1, \pm 2, \ldots\}$. $p_{i,\,i+1} = a$, $p_{i,\,i-1} = b$, $p_{ii} = c$, where $a + b + c = 1$, $a > 0$, $b > 0$, $c \geq 0$.

- Fix a state i.
- Suppose that $p_{ii}^{(1)} = c > 0$.
  - Then, $1 \in \left\{ n \geq 1 : p_{ii}^{(n)} > 0 \right\}$.
  - Hence, $d_i = 1$.
- Now, suppose that $p_{ii}^{(1)} = c = 0$.
  - Then, $p_{ii}^{(2m-1)} = 0$ and $p_{ii}^{(2m)} > 0$ for all $m \geq 1$.
  - Therefore, $\left\{ n \geq 1 : p_{ii}^{(n)} > 0 \right\} = \{2m : m \geq 1\}$.
  - Hence, $d_i = 2$.

Theorems

Theorem: If $i \leftrightarrow j$, then $d(i) = d(j)$.

Remark: Thus, if $\{X_n : n \geq 0\}$ is irreducible, then all states have same period. In particular, if all states have period 1, the MC is called aperiodic.

Theorem: Let $\{X_n : n \geq 0\}$ be an irreducible, positive recurrent, and aperiodic MC. Then, for any initial distribution μ, $\lim_{n\to\infty} P_\mu(X_n = i)$ exists and is equal to π_i for all i, where $(\pi_i : i \in S)$ is the unique stationary distribution. In particular, for any two states i and j, $\lim_{n\to\infty} p_{ij}^{(n)} = \pi_j$.

Problems

Example 4: Consider the MC with the following TPM

$$P = \begin{array}{c c} & \begin{array}{ccccc} 0 & 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix} \end{array}.$$

Find all stationary distributions.

Let $(\pi_0, \pi_1, \pi_2, \pi_3, \pi_4)$ be a stationary distribution. Then,

$$\pi_0 = \frac{1}{2}\pi_0 + \frac{1}{2}\pi_4 \qquad \pi_1 = \frac{1}{2}\pi_1 + \frac{1}{4}\pi_3$$

$$\pi_2 = \pi_2 + \frac{1}{4}\pi_3 \qquad \pi_3 = \frac{1}{2}\pi_1 + \frac{1}{4}\pi_3$$

$$\pi_4 = \frac{1}{2}\pi_0 + \frac{1}{4}\pi_3 + \frac{1}{2}\pi_4 \qquad \pi_0 + \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$$

Problems

$$\pi_0 = \frac{1}{2}\pi_0 + \frac{1}{2}\pi_4 \qquad \pi_1 = \frac{1}{2}\pi_1 + \frac{1}{4}\pi_3$$

$$\pi_2 = \pi_2 + \frac{1}{4}\pi_3 \qquad \pi_3 = \frac{1}{2}\pi_1 + \frac{1}{4}\pi_3$$

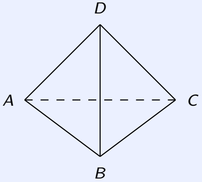
$$\pi_4 = \frac{1}{2}\pi_0 + \frac{1}{4}\pi_3 + \frac{1}{2}\pi_4 \qquad \pi_0 + \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$$

- From first equation, $\pi_0 = \pi_4$.
- From third equation, $\pi_3 = 0$.
- From fourth equation, $\pi_1 = 0$.
- Thus, $(\alpha, 0, \beta, 0, \alpha)$ is a stationary distribution if

$$0 \leq \alpha, \beta \leq 1 \quad \text{and} \quad 2\alpha + \beta = 1.$$

Problems

Example 5: A particle performs random walk on the vertices of a tetrahedron. At each step it remains where it is with probability 0.25 or moves to one of its neighboring vertices each with probability 0.25. Suppose that the particle starts at A (see the following figure). Find the mean number of steps until its first return to A.



- Let X_n be the position of the particle at time n. Then, $\{X_n : n \geq 0\}$ is a MC with TPM

$$P = \begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{bmatrix} \end{array}$$

Problems

- **Useful Fact:** An $n \times n$ TPM is said to be doubly stochastic if its' each column sum is also equal to 1. If $\{X_n : n \geq 0\}$ is an irreducible MC with state space S, with $|S| = n$ and a doubly stochastic TPM, then, the unique stationary distribution of the MC is given by $\left(\pi_i = \frac{1}{n} : i \in S\right)$.
- Using the above fact, the stationary distribution of $\{X_n : n \geq 0\}$ is $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$. Thus,

$$E_A(T_A) = \frac{1}{\frac{1}{4}} = 4.$$

Problems

Example 6: Each morning an individual leaves his house and goes for a run. He is equally likely to leave either from his front or back door. Upon leaving the house, he chooses a pair of running shoes (or goes running barefoot if there are no shoes at the door from which he departed). On his return, he is equally likely to enter, and leave his running shoes, either by the front or back door. If he owns a total of 4 pairs of running shoes, what is the long run proportion of time he runs barefooted?

Problems

- Let X_n be the number of pairs of shoes at the front door on the nth morning.
- Then, $\{X_n : n \geq 0\}$ is a MC with TPM

$$\begin{array}{c|c} & \begin{array}{ccccc} 0 & 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{4} & \frac{3}{4} \end{bmatrix} \end{array}.$$

- By Useful fact, the unique stationary distribution is

$$\left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}\right).$$

- Thus, the long run proportion of times he runs barefooted is

$$\frac{1}{2}\pi_0 + \frac{1}{2}\pi_4 = \frac{1}{5}.$$

Problems

Example 7: In a good weather year, the number of storms is Poisson distributed with mean 1. In a bad year, it is Poisson distributed with mean 3. Assume that any year's weather condition depends on past years only through the previous year condition. If a good year is equally likely to be followed either by a good year or a bad year, and that a bad year is twice as likely to be followed by a bad year as by a good year. Find the long run average number of storms per year.

- Let $X_n = 0$ if the nth year is bad and $X_n = 1$ if the nth year is good.
- Then, $\{X_n : n \geq 0\}$ is a MC with TPM $\begin{array}{c c} & \begin{array}{cc} 0 & 1 \end{array} \\ \begin{array}{c} 0 \\ 1 \end{array} & \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \end{array}$.
- The unique stationary distribution is $\left(\frac{3}{5}, \frac{2}{5}\right)$.
- Thus, the long run average number of storms per year is $\frac{3}{5} \times 3 + \frac{2}{5} \times 1 = \frac{11}{5}$.